

LAB SETUP

OVERVIEW:

This lab uses Kali Linux as the attacker and Ubuntu with DVWA as the victim, connected through an Internal Network in VirtualBox. It provides a isolated environment to learn and perform web vulnerability assessment based on the OWASP Top 10.

LAB ARCHITECTURE:

[Kali Linux] ---- Internal Network ---- [Ubuntu + DVWA]

Attacker

Victim Server

DOWNLOAD BEFORE CONNECTING TO INTERNAL NETWORK:

On Kali Linux

Pre-installed usually, but verify:

- Nmap
- Wireshark
- OWASP ZAP
- Firefox
- Wfuzz
- Nikto
- Gobuster

On Ubuntu:

- Apache
- MySQL
- PHP
- DVWA

SET UP BOTH KALI & UBUNTU :

- Update Kali Linux:
sudo apt update && sudo apt full-upgrade -y
- Verify Pre-Installed Tools in Kali
 - Nmap

nmap --version

- Wireshark

wireshark --version

- OWASP ZAP

zapproxy --version

- FIREFOX

firefox --version

- WFUZZ

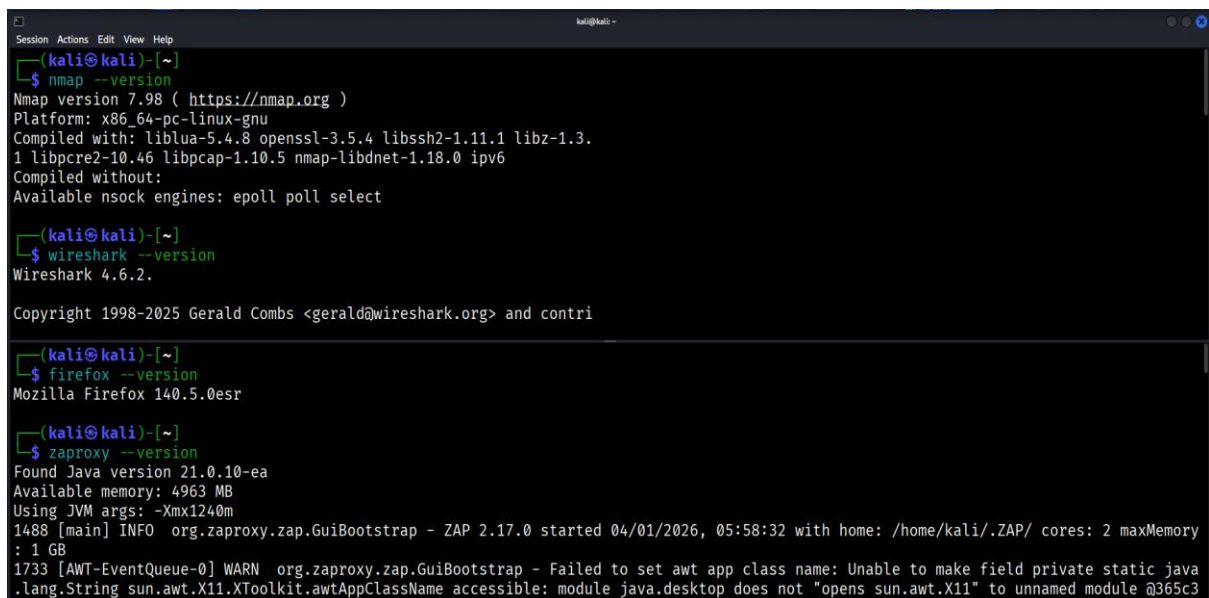
wfuzz --version

- GOBUSTER

gobuster --version

- NIKTO

dpkg -l | grep nikto



```
kali@kali:~$ nmap --version
Nmap version 7.98 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.5.4 libssh2-1.11.1 libz-1.3.
1 libpcap-1.10.5 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select

kali@kali:~$ wireshark --version
Wireshark 4.6.2.

Copyright 1998-2025 Gerald Combs <gerald@wireshark.org> and contri

kali@kali:~$ firefox --version
Mozilla Firefox 140.5.0esr

kali@kali:~$ zapproxy --version
Found Java version 21.0.10-ea
Available memory: 4963 MB
Using JVM args: -Xmx1240m
1488 [main] INFO org.zaproxy.zap.GuiBootstrap - ZAP 2.17.0 started 04/01/2026, 05:58:32 with home: /home/kali/.ZAP/ cores: 2 maxMemory
: 1 GB
1733 [AWT-EventQueue-0] WARN org.zaproxy.zap.GuiBootstrap - Failed to set awt app class name: Unable to make field private static java
.lang.String sun.awt.X11.XToolkit.awtAppClassName accessible: module java.desktop does not "opens sun.awt.X11" to unnamed module @365c3
```

```
(kali@kali)~$ dpkg -l | grep nikto
ii  nikto      1:2.5.0+git20230114.90ff645-0kali1  all          web server security scanner

(kali@kali)~$ gobuster --version
gobuster version 3.8
Build info:
go      go1.24.4
path    github.com/OJ/gobuster
build   -buildmode=exe
build   -compiler=gc
build   -trimpath=true
build   DefaultGODEBUG=asynctimerchan=1,gotestjsonbuildtext=1,gotypesalias=0,httplaxcontentlength=1,httpmuxgo121=1,httpservecontentkeep
headers=1,multipathhttp=0,netedns0=0,panicnil=1,randseednop=0,rsa1024min=0,tls10server=1,tls3des=1,tls1kcm=0,tlsrsakex=1,tlsunsafeekm=1
,winreadlinkvolume=0,winsymlink=0,x509keypairleaf=0,x509negativeserial=1,x509rsact=0,x509usepolicies=0
build   CGO_ENABLED=1
build   GOARCH=amd64
build   GOOS=linux
build   GOAMD64=v1

(kali@kali)~$ wfuzz --version
/usr/lib/python3/dist-packages/wfuzz/__init__.py:34: UserWarning:Pycurl is not compiled against Openssl. Wfuzz might not work correctl
y when fuzzing SSL sites. Check Wfuzz's documentation for more information.
3.1.0

(kali@kali)~$
```

UBUNTU – DVWA SERVER SETUP :

- UPDATE UBUNTU :
sudo apt update && sudo apt upgrade -y

```
vboxuser@UBUNTULINUX:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for vboxuser:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://in.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,391 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,684 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [225 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9,504 B]
Get:10 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,286 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [523 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [916 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [207 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [71.4 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [311 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/universe Icons (48x48) [46.6 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe Icons (64x64) [72.9 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.4 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [5,956 B]
Get:21 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:22 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:23 http://in.archive.ubuntu.com/ubuntu noble-updates/main Icons (48x48) [36.0 kB]
Get:24 http://in.archive.ubuntu.com/ubuntu noble-updates/main Icons (64x64) [51.0 kB]
Get:25 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.8 kB]
Get:26 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,413 kB]
Get:27 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [550 kB]
Get:28 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:29 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [516 B]
Get:30 http://in.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,506 kB]
Get:31 http://in.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [306 kB]
```

- INSTALL REQUIRED PACKAGES (LAMP) :

`sudo apt install apache2 mysql-server php php-mysql php-gd -y`

```
vboxuser@UBUNTULINUX:~$ sudo apt install apache2 mysql-server php php-mysql php-gd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libaio1t64 libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1
  -ldap libaprutil1t64 libcgi-fast-perl libcgi-pm-perl libevent-core-2.1-7t64 libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-p
  erl libfcgi0t64 libhtml-template-perl libmecab2 libprotobuf-lite32t64 libsodium23 mecab-ipadic mecab-ipadic-utf8 mecab-uti
  ls
  mysql-client-8.0 mysql-client-core-8.0 mysql-common mysql-server-8.0 mysql-server-core-8.0 php-common php8.3 php8.3-cl
  i
  php8.3-common php8.3-gd php8.3-mysql php8.3-opcache php8.3-readline
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom php-pear libipc-sharedcache-perl mailx tinycal
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libaio1t64 libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3 lib
  aprutil1-ldap
  libaprutil1t64 libcgi-fast-perl libcgi-pm-perl libevent-core-2.1-7t64 libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-p
  erl
  libfcgi0t64 libhtml-template-perl libmecab2 libprotobuf-lite32t64 libsodium23 mecab-ipadic mecab-ipadic-utf8 mecab-uti
  ls
  mysql-client-8.0 mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0 mysql-server-core-8.0 php php-common
  php-gd
  php-mysql php8.3 php8.3-cli php8.3-common php8.3-gd php8.3-mysql php8.3-opcache php8.3-readline
0 upgraded, 41 newly installed, 0 to remove and 0 not upgraded.
Need to get 36.5 MB of archives.
After this operation, 273 MB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9.11 kB]
```

- VERIFY APACHE:

`sudo systemctl status apache2`

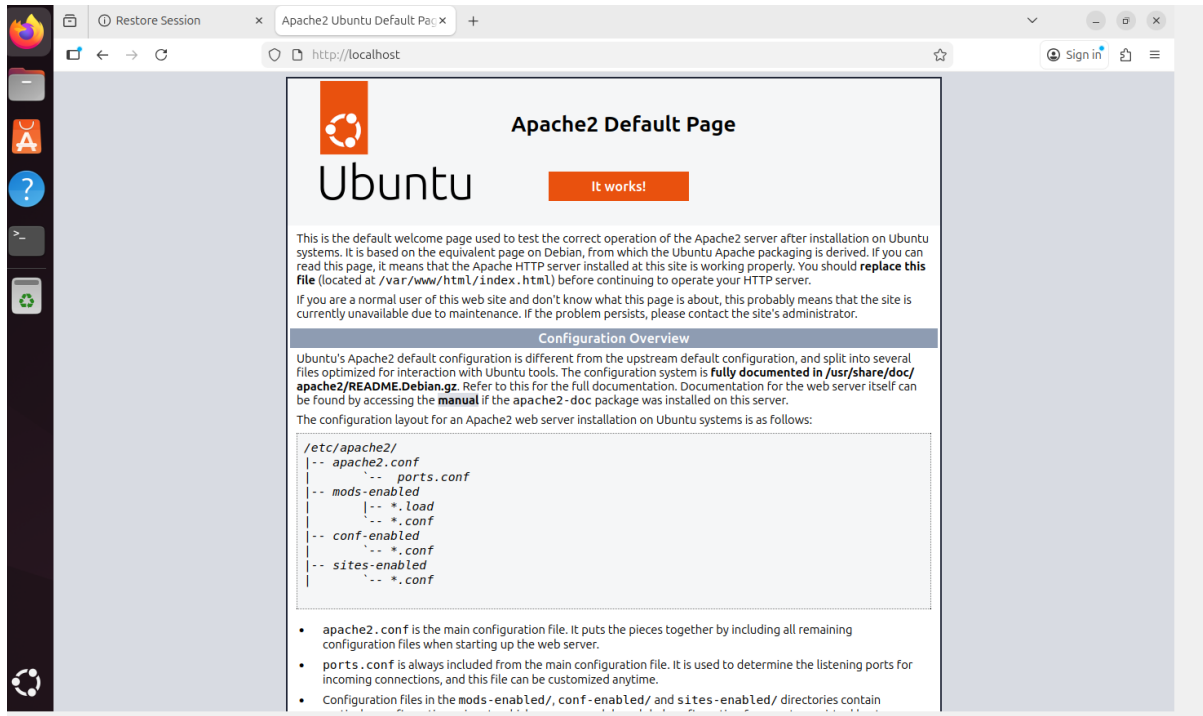
```
Processing triggers for php8.3-cli (8.3.6-0ubuntu0.24.04.5) ...
Processing triggers for libapache2-mod-php8.3 (8.3.6-0ubuntu0.24.04.5) ...
vboxuser@UBUNTULINUX:~$ sudo systemctl status apache2
[sudo] password for vboxuser:
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-12-31 06:36:34 UTC; 14min ago
     Docs: https://httpd.apache.org/docs/2.4/
  Process: 29961 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Main PID: 29966 (apache2)
       Tasks: 6 (limit: 7997)
      Memory: 11.5M (peak: 11.7M)
         CPU: 66ms
    CGroup: /system.slice/apache2.service
            └─29966 /usr/sbin/apache2 -k start
              └─29969 /usr/sbin/apache2 -k start
                └─29970 /usr/sbin/apache2 -k start
                  └─29971 /usr/sbin/apache2 -k start
                    └─29972 /usr/sbin/apache2 -k start
                      └─29973 /usr/sbin/apache2 -k start

Dec 31 06:36:34 UBUNTULINUX systemd[1]: Starting apache2.service - The Apache HTTP Server...
Dec 31 06:36:34 UBUNTULINUX apachectl[29965]: AH00558: apache2: Could not reliably determine the server's fully qualifie
d domain name;
Dec 31 06:36:34 UBUNTULINUX systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-20/20 (END)
[1]+  Stopped                  sudo systemctl status apache2
```

- TEST APACHE :

Open Ubuntu browser:

`http://localhost`



- DOWNLOAD DVWA :

`cd /var/www/html`

`sudo git clone https://github.com/digininja/DVWA.git`

Set permissions:

`sudo chown -R www-data:www-data DVWA`

`sudo chmod -R 755 DVWA`

```

Firefox
Setting up liberror-perl (0.17029-2) ...
Setting up git-man (1:2.43.0-1ubuntu7.3) ...
Setting up git (1:2.43.0-1ubuntu7.3) ...
Processing triggers for man-db (2.12.0-4build2) ...
vboxuser@UBUNTULINUX:/var/www/html$ sudo git clone https://github.com/digininja/DVWA.git
[sudo] password for vboxuser:
Cloning into 'DVWA'...
remote: Enumerating objects: 5622, done.
remote: Total 5622 (delta 0), reused 0 (delta 0), pack-reused 5622 (from 1)
Receiving objects: 100% (5622/5622), 2.64 MiB | 3.75 MiB/s, done.
Resolving deltas: 100% (2809/2809), done.
vboxuser@UBUNTULINUX:/var/www/html$ sudo chown -R www-data:ww-data DVWA
chown: invalid group: 'www-data:ww-data'
vboxuser@UBUNTULINUX:/var/www/html$ sudo chown -R www-data:www-data DVWA
vboxuser@UBUNTULINUX:/var/www/html$ sudo chmod -R 755 DVWA
vboxuser@UBUNTULINUX:/var/www/html$ cd DVWA/config
vboxuser@UBUNTULINUX:/var/www/html/DVWA/config$ sudo cp config.inc.php.dist config.inc.php
vboxuser@UBUNTULINUX:/var/www/html/DVWA/config$ sudo nano config.inc.php
vboxuser@UBUNTULINUX:/var/www/html/DVWA/config$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.44-0ubuntu0.24.04.2 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE dvwa;
Query OK, 1 row affected (0.12 sec)

```

- CONFIGURE DVWA DATABASE:

cd DVWA/config

sudo cp config.inc.php.dist config.inc.php

MySQL:

sudo systemctl status mysql

Must be:active(running)

To run sql:

PHP:

Check by opening :

http://127.0.0.1/

if apache page opens PHP works with apache

DVWA connection setting :

```
sudo nano /var/html/DVWA/config/config.inc.php
```

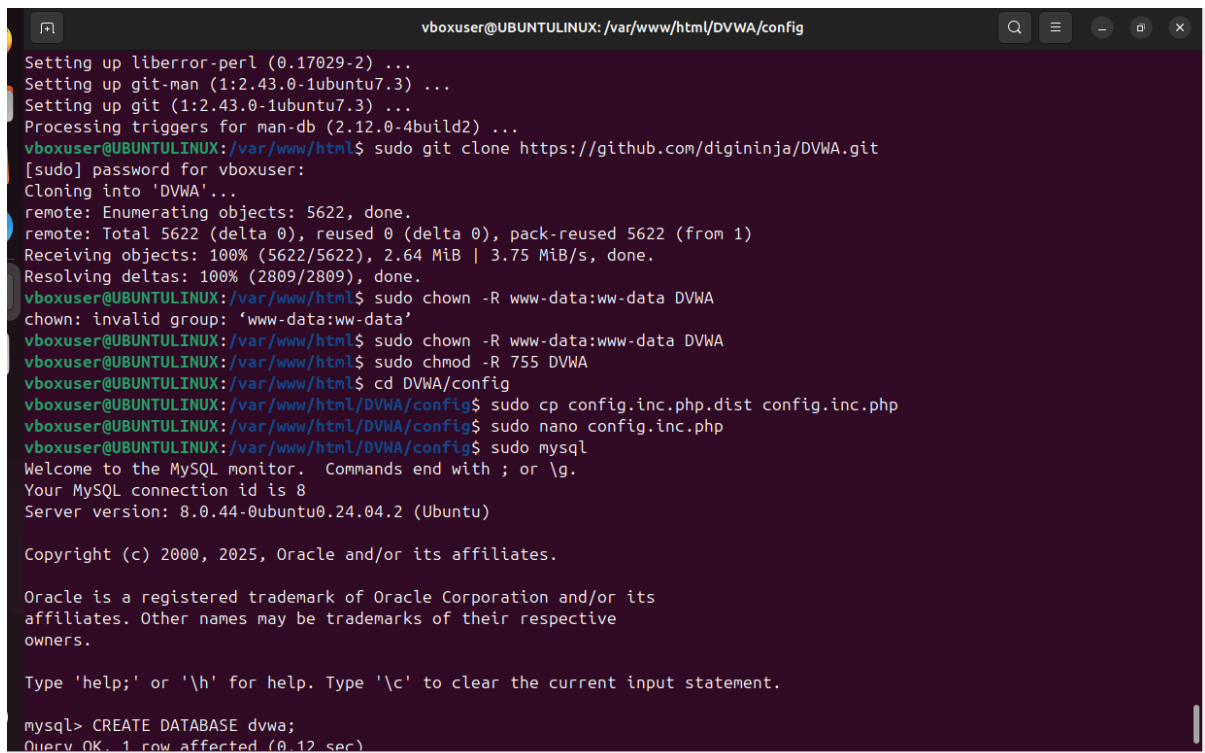
must be php code look like:

```
$_DVWA[ 'db_sever' ] = '127.0.0.1';
```

```
$_DVWA[ 'db_database' ] = 'dvwa';
```

```
$_DVWA[ 'db_user' ] = 'root';
```

```
$_DVWA[ 'db_password' ] = 'password';
```



```
vboxuser@UBUNTULINUX: /var/www/html/DVWA/config
Setting up liberror-perl (0.17029-2) ...
Setting up git-man (1:2.43.0-1ubuntu7.3) ...
Setting up git (1:2.43.0-1ubuntu7.3) ...
Processing triggers for man-db (2.12.0-4build2) ...
vboxuser@UBUNTULINUX:/var/www/html$ sudo git clone https://github.com/digininja/DVWA.git
[sudo] password for vboxuser:
Cloning into 'DVWA'...
remote: Enumerating objects: 5622, done.
remote: Total 5622 (delta 0), reused 0 (delta 0), pack-reused 5622 (from 1)
Receiving objects: 100% (5622/5622), 2.64 MiB | 3.75 MiB/s, done.
Resolving deltas: 100% (2809/2809), done.
vboxuser@UBUNTULINUX:/var/www/html$ sudo chown -R www-data:www-data DVWA
chown: invalid group: 'www-data:ww-data'
vboxuser@UBUNTULINUX:/var/www/html$ sudo chown -R www-data:www-data DVWA
vboxuser@UBUNTULINUX:/var/www/html$ sudo chmod -R 755 DVWA
vboxuser@UBUNTULINUX:/var/www/html$ cd DVWA/config
vboxuser@UBUNTULINUX:/var/www/html/DVWA/config$ sudo cp config.inc.php.dist config.inc.php
vboxuser@UBUNTULINUX:/var/www/html/DVWA/config$ sudo nano config.inc.php
vboxuser@UBUNTULINUX:/var/www/html/DVWA/config$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.44-0ubuntu0.24.04.2 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE dvwa;
Query OK, 1 row affected (0.12 sec)
```

How to verify apache < -- > PHP < -- > MySQL connection:

Test mysql :

```
Mysql -u root -p
```

Password: // use mysql password here //

```

Query OK, 0 rows affected (0.06 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.05 sec)

mysql> EXIT;
Bye
vboxuser@UBUNTULINUX:~$ sudo systemctl restart mysql
vboxuser@UBUNTULINUX:~$ mysql -u root -p
Enter password:
ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: YES)
vboxuser@UBUNTULINUX:~$ mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.44-0ubuntu0.24.04.2 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> EXIT;
Bye
vboxuser@UBUNTULINUX:~$ php -m | grep mysqli
mysqli
vboxuser@UBUNTULINUX:~$ sudo apt install php-mysqldb -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package php-mysqldb
vboxuser@UBUNTULINUX:~$ sudo systemctl restart apache2
vboxuser@UBUNTULINUX:~$ sudo systemctl restart mysql
vboxuser@UBUNTULINUX:~$

```

If login works mysql ok. Sometimes an error appears when trying to log in: “Access denied for user ‘root’@‘localhost’.”

To solve this:

Step1: login using sudo

Sudo mysql

If you see:

mysql>

It is correct.

Step2: check how root is authenticated

Inside mysql :

```
SELECT user, host, plugin FROM mysql.user WHERE user='root';
```

You will see:

Plugin=auth_socket

That is the problem.

Step3: change root to password authentication

Inside mysql run :

```
ALTER USER 'root'@'localhost' IDENTIFIED WITH
mysql_native_password BY 'password';
```

```
FLUSH PRIVILEGES;
```

```
EXIT;
```

After this, if an error appears stating that the password does not satisfy the current policy requirements, the issue can be resolved by setting a strong password in MySQL.

```
ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY
'P@ssw0rd123';
```

```
FLUSH PRIVILEGES;
```

```
mysql> EXIT;
Bye
vboxuser@UBUNTU19:~$ sudo systemctl restart mysql
vboxuser@UBUNTU19:~$ mysql -u root -p
Enter password:
ERROR 1698 (28000): Access denied for user 'root'@'localhost'
vboxuser@UBUNTU19:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.44-0ubuntu0.24.04.2 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SELECT user, host, plugin FROM mysql.user WHERE user='root';
+-----+-----+-----+
| user | host      | plugin      |
+-----+-----+-----+
| root | localhost | auth_socket |
+-----+-----+-----+
1 row in set (0.01 sec)

mysql> ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY 'password@123';
ERROR 1819 (HY000): Your password does not satisfy the current policy requirements
mysql> ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY 'p@ssw0rd123';
ERROR 1819 (HY000): Your password does not satisfy the current policy requirements
mysql> ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY 'P@ssw0rd123';
Query OK, 0 rows affected (0.06 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.05 sec)
```

Step4: restart mysql

```
sudo systemctl restart mysql
```

step5: test login again

```
mysql -u root -p
```

enter password

Then restart both Apache and MySQL, and try to access DVWA in the browser. If an error appears, such as:

SQL syntax error near 'IF NOT EXISTS role VARCHAR(20) DEFAULT "user"'

Run:

```
mysql --version
```

version comes

Then,

Step 1: Open the DVWA MySQL file

```
sudo nano /var/www/html/DVWA/dvwa/includes/DBMS/MySQL.php
```

Step 2: Find this line (around line 68)

```
$sql = "ALTER TABLE users ADD COLUMN IF NOT EXISTS role VARCHAR(20)
DEFAULT 'user'";
```

Step 3: Replace it with this,

```
$sql = "ALTER TABLE users ADD COLUMN role VARCHAR(20) DEFAULT 'user'";
```

Step 4: Save & Exit

Step 5: Restart services

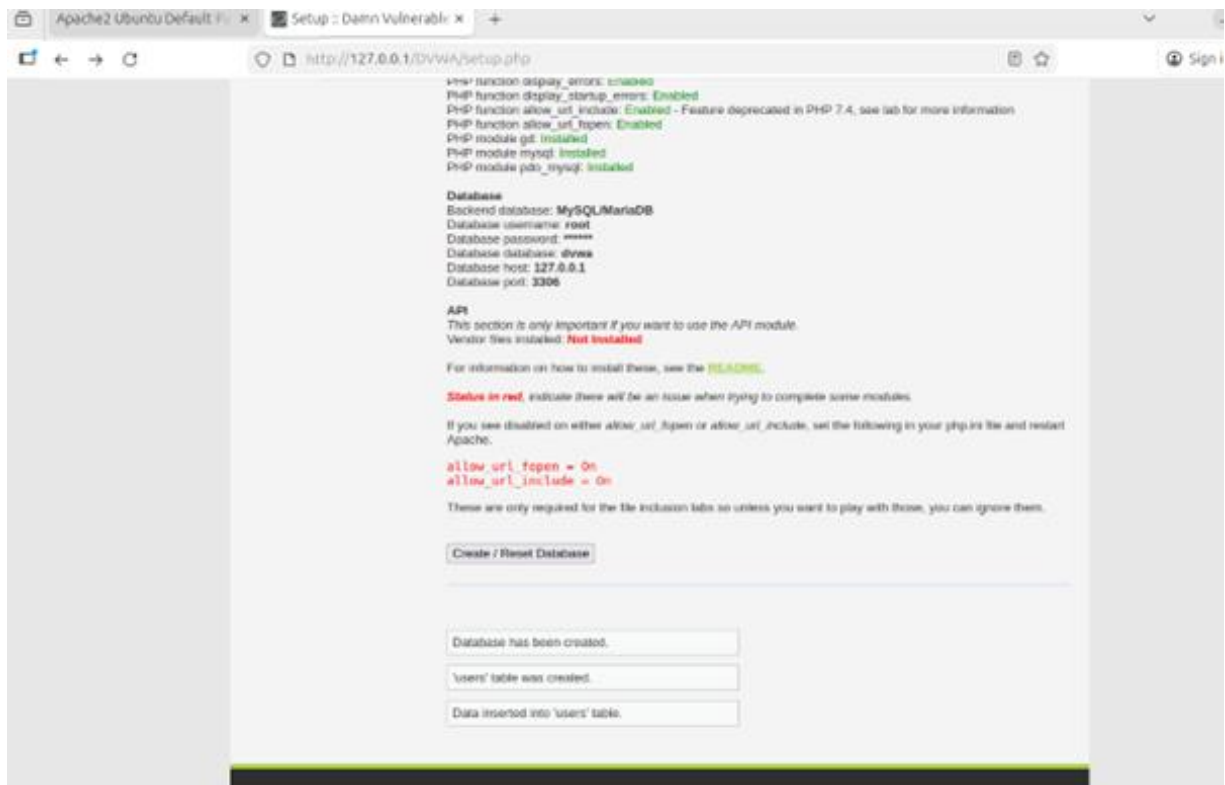
```
sudo systemctl restart mysql
```

```
sudo systemctl restart apache2
```

Step 6: Now open DVWA setup again

Browser:

```
http://127.0.0.1/DVWA/setup.php
```



LOGIN TO DVWA:

<http://<UBUNTU-IP>/DVWA/login.php>

Default credentials:

username: admin

password: password

Go to:

DVWA Security → set to LOW

Lab is ready

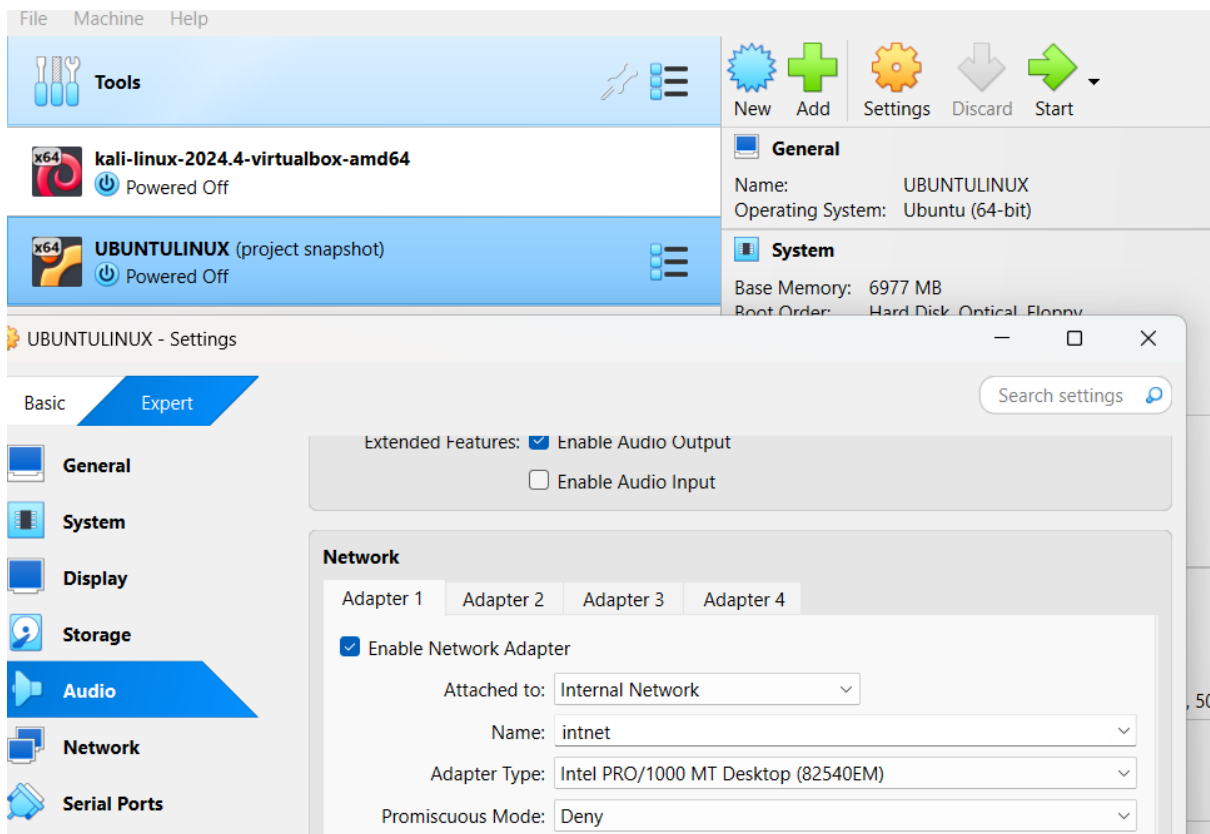
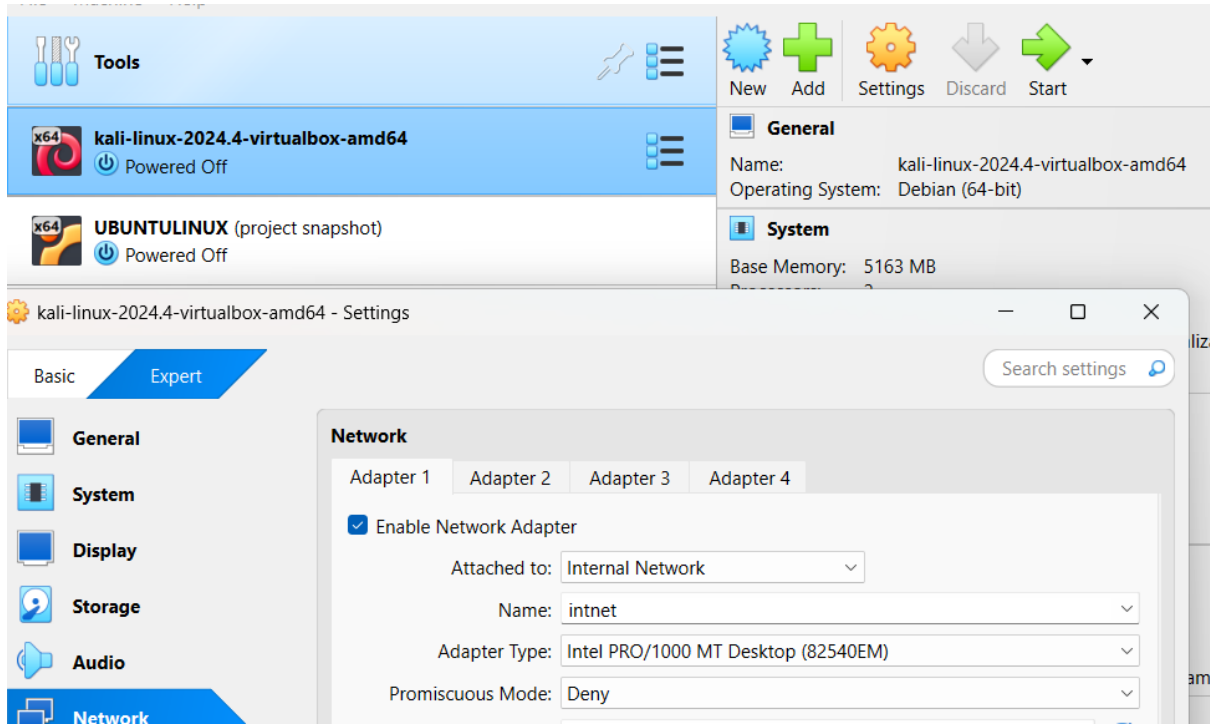
VIRTUALBOX NETWORK SETUP :

For both Kali & Ubuntu,

Open VirtualBox:

- Select VM → Settings
- Go to Network

- Adapter 1:
- Attached to: Internal Network
- Name: intnet (same for both VMs)
- Click OK



ASSIGNING STATIC IP ADDRESS :

For Kali Linux:

- Find interface

```
ip a
```

- Set static IP (CLI only)

```
sudo nmcli con show
```

Find connection name (usually Wired connection 1)

- Assign static IP

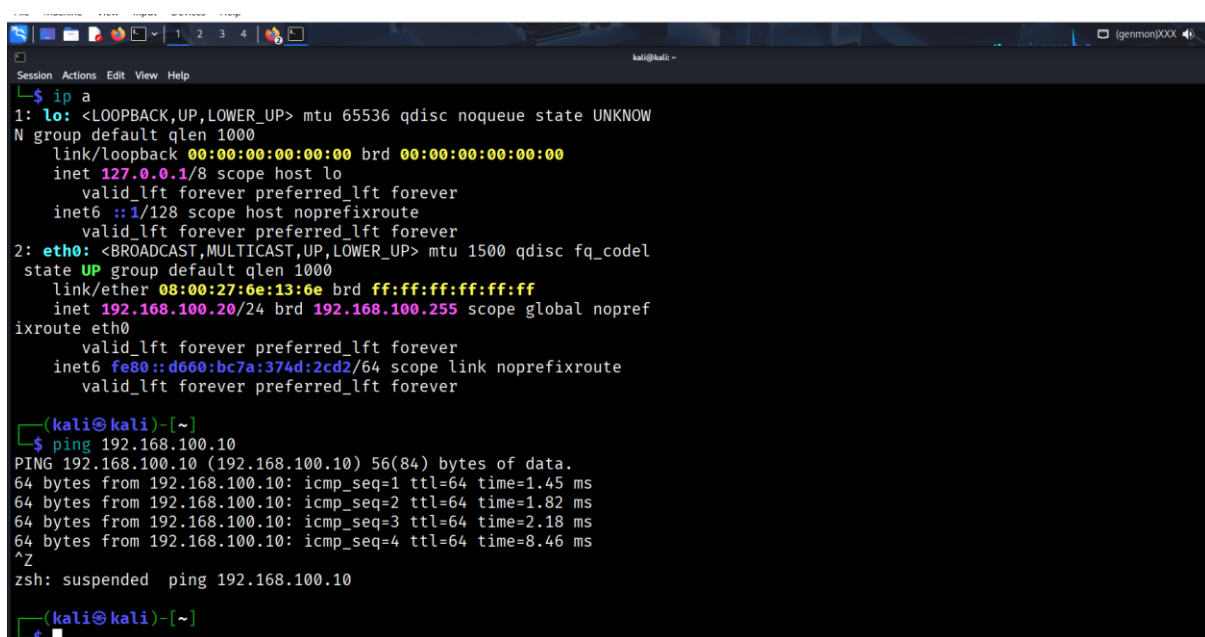
```
sudo nmcli con mod "Wired connection 1" ipv4.method manual
```

```
sudo nmcli con mod "Wired connection 1" ipv4.addresses 192.168.100.20/24
```

```
sudo nmcli con up "Wired connection 1"
```

- Verify

```
ip a show eth0
```



```
kali@kali: ~  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel  
    state UP group default qlen 1000  
    link/ether 08:00:27:6e:13:6e brd ff:ff:ff:ff:ff:ff  
    inet 192.168.100.20/24 brd 192.168.100.255 scope global nopref  
ixroute eth0  
        valid_lft forever preferred_lft forever  
        inet6 fe80::d660:bc7a:374d:2cd2/64 scope link noprefixroute  
            valid_lft forever preferred_lft forever  
  
(kali@kali)~  
$ ping 192.168.100.10  
PING 192.168.100.10 (192.168.100.10) 56(84) bytes of data.  
64 bytes from 192.168.100.10: icmp_seq=1 ttl=64 time=1.45 ms  
64 bytes from 192.168.100.10: icmp_seq=2 ttl=64 time=1.82 ms  
64 bytes from 192.168.100.10: icmp_seq=3 ttl=64 time=2.18 ms  
64 bytes from 192.168.100.10: icmp_seq=4 ttl=64 time=8.46 ms  
^Z  
zsh: suspended ping 192.168.100.10  
  
(kali@kali)~  
$
```

For ubuntu:

- Find interface

```
ip a
```

- Edit Netplan (CLI)

```
sudo nano /etc/netplan/01-network-manager-all.yaml
```

- Static IP config

```
network:
```

```
version: 2
```

```
renderer: NetworkManager
```

```
ethernets:
```

```
enp0s3:
```

```
dhcp4: no
```

```
addresses:
```

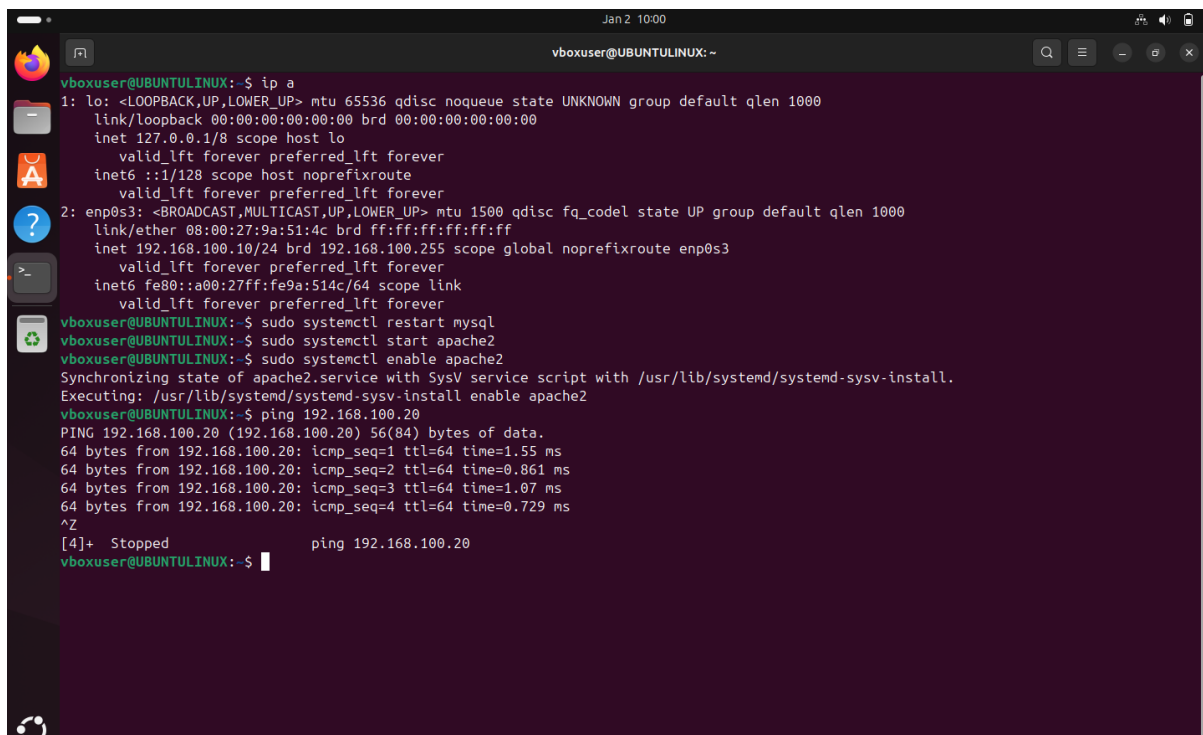
```
- 192.168.100.10/24
```

- Apply

```
sudo netplan apply
```

- Verify

```
ip a show enp0s3
```

A terminal window titled 'vboxuser@UBUNTULINUX: ~' showing network configuration and service management. The user runs 'ip a' to show details for 'lo' and 'enp0s3'. Then, they run 'sudo systemctl restart mysql', 'sudo systemctl start apache2', and 'sudo systemctl enable apache2'. A message indicates the service is being synchronized with SysV. Finally, they run 'ping 192.168.100.20', showing successful ping results. The terminal output is as follows:

```
vboxuser@UBUNTULINUX:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9a:51:4c brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.10/24 brd 192.168.100.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe9a:514c/64 scope link
        valid_lft forever preferred_lft forever
vboxuser@UBUNTULINUX:~$ sudo systemctl restart mysql
vboxuser@UBUNTULINUX:~$ sudo systemctl start apache2
vboxuser@UBUNTULINUX:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
vboxuser@UBUNTULINUX:~$ ping 192.168.100.20
PING 192.168.100.20 (192.168.100.20) 56(84) bytes of data.
64 bytes from 192.168.100.20: icmp_seq=1 ttl=64 time=1.55 ms
64 bytes from 192.168.100.20: icmp_seq=2 ttl=64 time=0.861 ms
64 bytes from 192.168.100.20: icmp_seq=3 ttl=64 time=1.07 ms
64 bytes from 192.168.100.20: icmp_seq=4 ttl=64 time=0.729 ms
^Z
[4]+  Stopped                  ping 192.168.100.20
vboxuser@UBUNTULINUX:~$
```

CHECK BOTH MACHINES CONNECTS TOGETHER OR NOT:

in kali : ping 192.168.100.10 #ubuntu ip

in ubuntu: ping 192.168.100.20 # kali ip

In case the ping request fails, the Apache service status is verified and restarted if required:

```
sudo systemctl status apache2
```

if it inactive then ok, else stop apache and restart it by:

```
sudo systemctl start apache2
```

```
sudo systemctl enable apache2
```

